

RADAR: Radiology AI-Driven Automated Reporting

AI-Assisted Diagnostic Imaging Report · B.Tech Computer Science Engineering · BVRIT Hyderabad (BVRIT Narsapur)

NAME:	—	PATIENT NUMBER:	UPLOAD_081011
REF. PHYSICIAN:	RADAR AI System	STUDY DATE:	11 May 2026
EXAM:	CT SCAN	STATUS:	CRITICAL

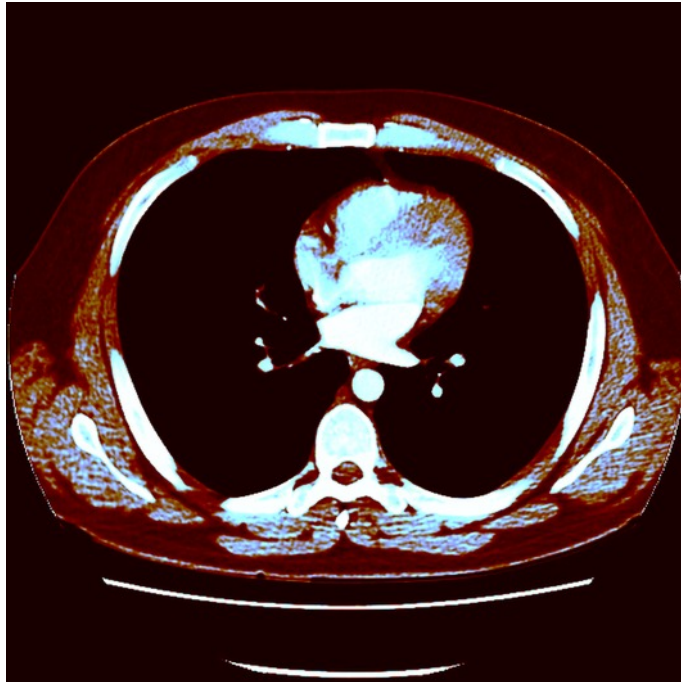


Figure 1. CT SCAN — AI-preprocessed input scan. 3-channel HU windowing: R=Bone/Lung (WW2250/WL-100), G=Soft Tissue (WW350/WL40), B=Brain (WW80/WL40). Slice shown: mid-volume. 896×896 px.

PROCEDURE:

Axial CT images of the chest obtained at 1.25mm intervals. No contrast.

COMPARISON:

COMPARISON:

None available.

FINDINGS:

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Lung parenchyma: There is a 1.8 cm spiculated nodule in the right upper lobe. No pleural effusion or pneumothorax identified. The left lung appears clear with no obvious masses or consolidations noted.

Mediastinum: The mediastinum demonstrates unremarkable contours and no significant lymphadenopathy is apparent on these non-contrast images.

Heart: The cardiac silhouette is within normal limits in size and configuration.

Osseous structures: No acute osseous abnormalities are seen.

IMPRESSION:

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1. Right Upper Lobe Spiculated Nodule - Suspicion for malignancy warrants further evaluation via tissue sampling.

RECOMMENDATIONS:

RECOMMENDATIONS:

- 1. Consider CT-guided biopsy of the right upper lobe nodule.
- 2. If biopsy confirms malignancy, consider PET/CT for staging purposes.

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■ CRITICAL FINDING — IMMEDIATE CLINICAL ATTENTION REQUIRED
No pleural effusion or pneumothorax identified.

AI PIPELINE METRICS

RAMT Confidence	59.7% [MED]	NLI Faithfulness	22.5% [LOW]	Report Quality	76.7% [HIGH]
Word Count	130	Inference Time	54.6 s	Urgency	CRITICAL

GHFE Detection Results (Top 5):

Finding	Probability	Level
Lung Mass	0.9959	HIGH
Mediastinal Adenopathy	0.9953	HIGH
Consolidation	0.9944	HIGH
Pleural Effusion	0.9939	HIGH
Hemorrhagic Stroke	0.9920	HIGH

RAGAS Quality Metrics:

Metric	Score
Faithfulness	0.8125 [HIGH]
Answer Relevancy	0.4287 [LOW]
Context Recall	0.1707 [LOW]
Context Precision	0.8571 [HIGH]
Hallucination Rate	0.1875 [LOW]
Contradiction Rate	0.1875 [LOW]
Overall Score	0.5971 [MED]

NLI Faithfulness Audit — Flagged Sentences:

- The left lung appears clear with no obvious masses or consolidations noted.
- 2.
- ...

RESEARCH PROTOTYPE — FOR ACADEMIC PURPOSES ONLY

This report was generated automatically by RADAR (Radiology AI-Driven Automated Reporting), a B.Tech Computer Science Engineering project at BVRIT Hyderabad (BVRIT Narsapur). Pipeline: DenseNet-121 GHFE → RAMT → RadioShield RAG → MedGemma 4B. This output has NOT been clinically validated and MUST NOT replace a licensed radiologist.

Case: UPLOAD_081011 | CT SCAN | 11 May 2026 08:11 | RADAR v3.1